



Hub & Spoke Design

Manufacturing ERP System

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Client: Manufacturer

Estimated Duration: ~3 Weeks

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1) Executive Summary

This document details the hub-spoke network design for the Manufacturing ERP System across two Azure regions: Italy North (Primary) and North Europe (DR). The architecture is based on Azure Well-Architected Framework principles and best practices to achieve a scalable, secure, and compliant network topology, using Zero Trust principles, Azure Firewall, and WAF for centralized security enforcement.

2) Network Topology Overview

High-Level Architecture

The **Hub-and-Spoke** model is employed with **DR** between regions:

- **Hub VNet:** Acts as the central point for networking, security, and VPN gateways.
- **Spoke VNets:**
 - **Spoke-DB:** SQL Server and database resources
 - **Spoke-App:** Application tier (IIS Web Servers)
 - **Spoke-Storage:** Private endpoints for storage

The primary region (Italy North) is mirrored in the DR region (North Europe).



Design Highlights:

- **Security Model:** Azure Firewall at the center for traffic filtering
- **Scalability:** /16 address spaces for future expansion
- **DR Readiness:** Mirrored architecture in the DR region
- **VNet Peering:** Ensures seamless connectivity between hub and spokes

3) Primary Region - Italy North

Hub Virtual Network

- **VNet Name:** vnet-erp-hub-italynorth
- **Address Space:** 10.0.0.0/16
- **Purpose:** Centralized connectivity, security (via Azure Firewall, VPN Gateway)

Subnets in Hub VNet

Subnet Name	CIDR	Usable IPs	Purpose
GatewaySubnet	10.0.1.0/27	27	VPN Gateway
AzureFirewallSubnet	10.0.2.0/26	59	Azure Firewall



Resources in Hub VNet

- **Azure Firewall** (fw-erp-italynorth))
- **VPN Gateway** (vpngw-erp-italynorth)

4) Spoke Virtual Networks

Spoke-DB (SQL Tier)

- **VNet Name:** vnet-erp-spoke-db-italynorth
- **Address Space:** 10.1.0.0/16
- **Subnets:**
 - **snet-sql:** 10.1.1.0/24 (SQL VMs, NSG: nsg-sql)

Spoke-App (Web Tier)

- **VNet Name:** vnet-erp-spoke-app-italynorth
- **Address Space:** 10.2.0.0/16
- **Subnets:**
 - **snet-web:** 10.2.1.0/24 (IIS Web Servers, NSG: nsg-web)



Spoke-Storage (Private Endpoints)

- **VNet Name:** vnet-erp-spoke-storage-italynorth
- **Address Space:** 10.3.0.0/16
- **Subnets:**
 - **snet-storage-pe:** 10.3.1.0/24 (Private Endpoints, NSG: nsg-pe)

5) DR Region - North Europe

DR Hub Virtual Network

- **VNet Name:** vnet-erp-hub-dr-northeurope
- **Address Space:** 10.10.0.0/16
- **Purpose:** DR Centralized Connectivity

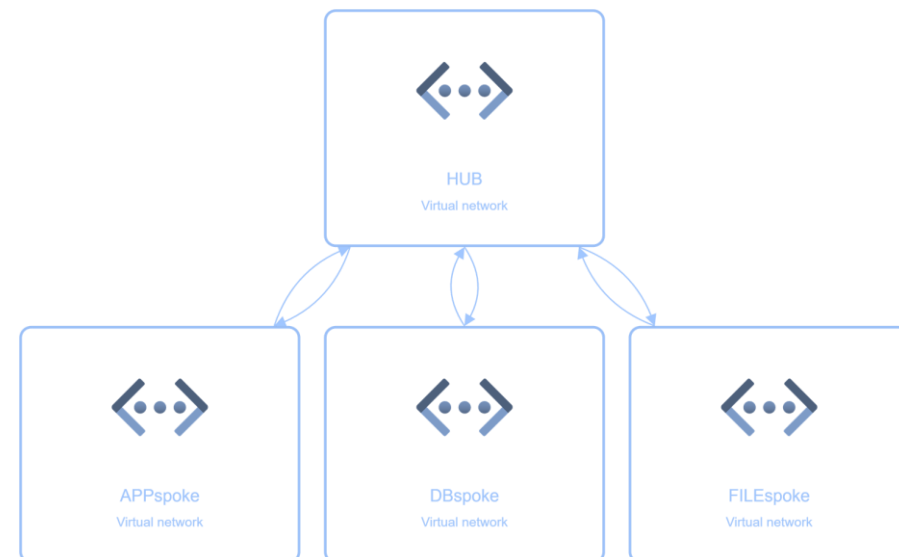


Subnets in DR Hub

Subnet Name	CIDR	Usable IPs	Purpose
GatewaySubnet	10.10.1.0/27	27	Future VPN (if needed)
AzureFirewallSubnet	10.10.2.0/26	59	DR Azure Firewall

6) VNet Peering Configuration

- Hub ↔ App Spoke (VNet peering)
- Hub ↔ DB Spoke (VNet peering)
- Hub ↔ Storage Spoke (VNet peering)
- No direct spoke-to-spoke peering (traffic forced via Hub/Firewall)





7) Traffic Flow Matrix

- Inbound Traffic: Traffic from Internet to Azure Front Door, then routed to Firewall then to Web VM.
- App → DB: Web tier connects to SQL VM via Firewall for traffic filtering (SQL port 1433).
- VPN Client Traffic: Admin VPN clients access via VPN Gateway, enforced by NSGs and Firewall.
- Outbound Internet Traffic: Routed through Azure Firewall.

8) Security Design

- Zero Trust Model: NSGs and Azure Firewall are implemented at every layer for traffic control.
- NSG Best Practices: Role-based rules for SQL, Web.
- Firewall Rules: DNAT and Network Rules filter traffic based on source, destination, and protocol.

9) Architecture Diagram

